Task 2 - Compare the features and functions of different operating systems

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Functions of Windows 7 & 10 Operating Systems

Kernel

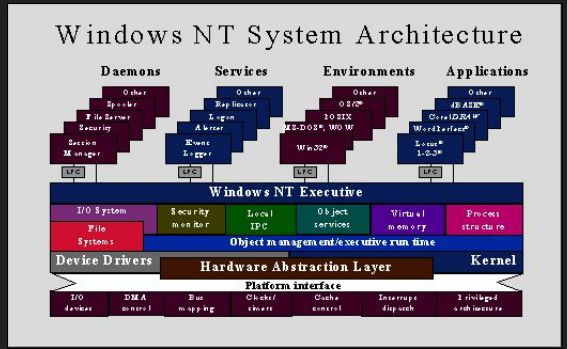
Windows 7

Windows 7 uses the Windows NT Kernel.

Windows 10

Windows 10 also uses the Windows NT Kernel.

Technically, there is no difference of the kernel between windows 10 and windows 7. The kernel in both windows are both the same kernels because they both contain the Windows NT Kernel, which has been used in Windows since Windows NT. The versions that the Windows NT kernel has been used on is: Windows NT 3.1, NT 3.5, NT 3.51, NT 4.0, 2000, XP, VIsta, 7, 8, 8.1 and 10.



User Interface

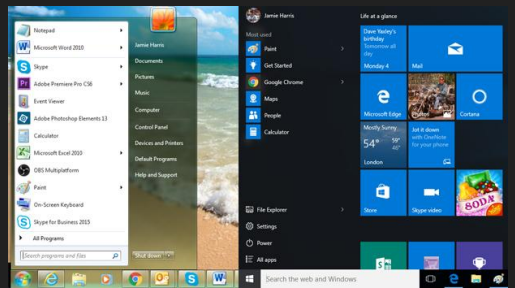
Windows 7

On Windows 7 you'll be used to clicking the Start Button and searching for content on your PC that way.

Windows 10

On Windows 10, it's all about Cortana, Microsoft's digital assistant which began life on Windows Phone.

The overall design of Windows 10 is the most striking difference to Windows 7. Everything has become flat, colourful and squared, unlike Windows 7 which was a bit glassy, rounded and used some gradients. Icons have also been simplified, but not beyond the point of recognition, again swapping the rounded look for something flat. With Windows 10, the search bar moved into the task bar with cortana added, instead of the start menu on Windows 7.



Device Drivers

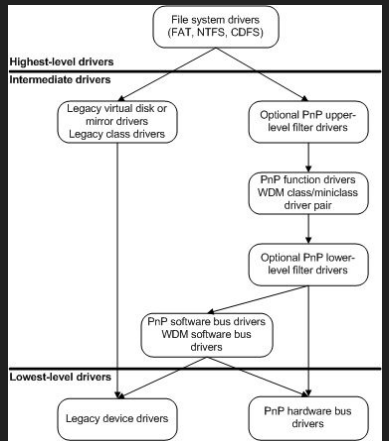
Windows 7

Windows 7 has a version of Microsoft WIndows Drivers that, if it hasn't been updated, would be outdated today

Windows 10

Since Windows 10 is a fairly new OS, it’s drivers are much more up to date than WIndows 7, but may still require updates.

Both drivers on the WIndows OS are the same type of drivers, so technically, there is no difference between the two. The only actual difference is the version of the driver to support the hardware that was popular of the time of the update.



File System Management

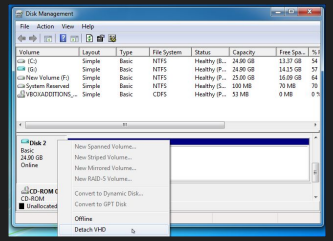
Windows 7

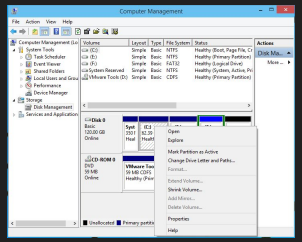
Windows 7 has a Disk manager with a glassy theme

Windows 10

Windows 10 has a Computer Management with a flat theme

There isn't much difference between WIndows 7 and Windows 10, but there are a few differences. Windows 7 has a different name of “Disk Manager” compared to Windows 10 “Computer Management”. Another difference is that Windows 10 has 4 different sections on the window compared to the 2 different sections in the windows 7. In Windows 10, the panel on the left is the navigation throughout the disk, which Windows 7 doesn’t have. The panel on the right on Windows 10 has actions which is used for more actions for the computer management, which Windows 7 doesn’t also have.





Registry

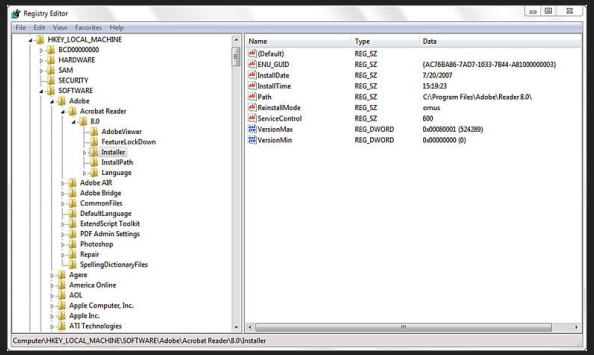
Windows 7

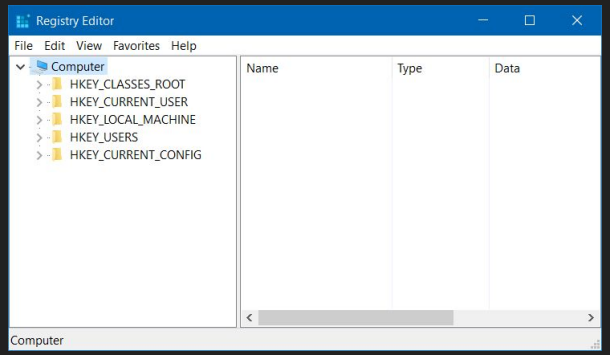
Has a Registry editor with the Windows 7 glassy theme

Windows 10

Has a Registry editor with the Windows 10 flat theme

The registry on both windows are also very similar visually because it is a very simplistic design that doesn't need any more additions, but there is a different aesthetic to both of them. The Windows 10 Registry has the “flat” theme which Windows 10 theme has, compared to the WIndows 7 “glassy” theme, but that is the only difference between the two.





Features of Windows 7 & 10 Operating Systems

Customizing the User Interface (Desktop, etc.)

Windows 7

With Windows 7, there is no way of heavily customising the UI like Windows 10, if there is any customisation at all. This is why Windows 7 looks similar on all machines, compared to the mass difference in other windows 10 machines.

Windows 10

As I mentioned in the last powerpoint, with windows 10, you can heavily customize the user interface better than the previous Windows OS. This can give you the ability to personalise it to your taste.

With workspace management, Windows 7 has multi-monitor support, though it’s somewhat limited with just access to either a mirroring mode or one that stretches the whole desktop across all the monitors, but with only one taskbar on one monitor. IT also has an early version of snap, the tool that automatically resizes windows when you drag them to the left or right.

With Windows 10 both these features have been enhanced. Different monitors can now have different backgrounds, with different slideshows too. Pictures can also now be set to span across all the monitors.

File Management

Windows 7

Windows 7 has a decent selection and certainly many people are used to them. However, they certainly aren’t the most advanced, and in some ways they’re a backwards step from what came before.

This one little feature summed up all that was wrong about the nonsensical path being trodden by Microsoft at the time. However, it wasn’t the only problem with Windows 7 File explorer. By any standard it was poorly designed, with wasted space and key functions hidden from view.

Windows 10

Windows 10 File Explorer and copy dialogue is much more useful, especially with addition in storage spaces. THis is a way to group multiple hard drives together to form a single logical drive.

While hardly revolutionary sounding, what makes it particularly useful is that it supports multiple levels of drive redundancy, to ensure that the data is backed up, and unlike a RAID system it’s completely hot-swappable and drives can be different sizes

Application Management

Windows 7

With Windows 7, there is only 1 way to manage applications, which is through the settings window.

This may seem like a good enough way of managing apps, but when compared to Windows 10, it cannot be accessed as easily or as versatile as Windows 10 is.

Windows 10

With Windows 10, there are many ways of managing applications. Which is via start menu, control panel and the settings window.

This makes it a lot easier for the user to see the apps and what the user can do with the apps depending on which method he is using to access the applications.

Basically, Windows 10 is much more faster for application management. For example, if you wanted to delete an application, the time it would take to do in Windows 10 would be significantly faster than Windows 7. This is due to the fact that you can do it just from the start menu in Windows 10 and for Windows 7 you needed to go to through the control panel to find the application management tool.

System Management Utilities

Windows 7

Windows 7 opens a window called “System Configuration”

Windows 10

Windows 10 also opens a window called “System Configuration”

With both Operating Systems, the MSCONFIG is very similar with the only difference being the theme. With both OS, it does the exact same thing of disabling and re-enabling software, device drivers or Windows services that run during startup process to help the user determine the cause of a problem with Windows.

The reason they are the same is that it only varies in Windows older than Vista. WIth VIsta and above, the tool gained additional support for launching a variety of tools, such as system information, other configuration areas, such as Internet options, and the ability to enable/disable UAC. Basically, there are no differences between the two.

Windows Maintenance Tasks

Windows 7

With Windows 7, It is opened via the control panel.

Windows 10

With Windows 10, It is also opened via the control panel.

As I mentioned before, Windows 10 has improved with the tools to make it a lot easier to take care of the PC. The automatic maintenance in WIndows 10 is the maintenance task on a windows 10 PC. What I meant by this is that Windows 10 has improved from the previous Windows OS because the PC can perform an automatic maintenance, which before you had to manually go through a wizard, like in windows 7, to choose what kind of maintenance you want your PC to carry out.

This means that a PC user that is less experienced would find Windows 10 much more advantageous for this because the PC does the work for you instead of you having to find out what kind of maintenance your PC needs.

Strengths and weaknesses

There’s nothing wrong with sticking with Windows 7, but upgrading to Windows 10 definitely has plenty of benefits, and not too many downsides.

Windows 10 has a lot of genuinely useful new features, as well as key improvements to old features, compared to Windows 7. None is enough on its own to make an upgrade necessary, but when combined they do make a compelling case. You’ve got better search, window management, file management, tools and more. Windows 10 is faster in general use too, and the new Start Menu is in some ways better than the one in Windows 7.

A downside to Windows 10 that Windows 7 doesn't have however, is not having control over Windows updates. Windows 10 completely removes the ability to manually control your system updates. Windows 10 Pro will let you defer updates for a few months, but that’s the extent of it. This isn’t much of a big deal for people who are stuck with Windows 7’s default Windows Update options, which downloaded patches automatically. But if you like to control your own update destiny, this could be a showstopper.